MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 11745212
BITE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Bite.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11745212	Circuit Card Assembly, Bite

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition.- This assembly is a single ended printed circuit board consisting of op-amps, gates, word selectors, diodes, counters, and a flip flop necessary to provide timing and bit signals for the built-in test section of the STG.

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3.1.1 Major component list. - The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11745212.

3.2 Characteristics.

3.2.1 Performance. - The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power Supply. - The requirements of 3.2 shall be met with the following supplies:

a. +5 VDC
b. +15 VDC
c. +28 VDC
d. -5 VDC
e. -15 VDC

3.2.1.2 Voltage divider/Nand gate network. - Inputs are applied to PINS 36 & 29 and outputs are yielded on PINS 36 & 37 as shown in Table 1.

3.2.1.3 Flip flop/Nand gate network. - Inputs are applied to PINS 39, 29, 23, & 32 and the output is yielded on PIN 41 as shown in Table 2.

3.2.1.4 Nand gate. - The input is applied to PIN 25, the other input is permanently high, and the output is yielded on PIN 13 as shown in Table 3.

3.2.1.5 Flip flop/Nand gates/Decade counter network. - Inputs are applied to PINS 23, 17, & 31 and the output is yielded on PIN 33.

3.2.1.6 4-Bit data selector/Storage register #1. - Four of the selector/storage register inputs are applied to PINS 19, 20, 21 & 27 and the other four are received internally from a decade counter; the outputs are yielded on PINS 22, 24, 26, & 28.

3.2.1.7 4-Bit data selector/Storage register #2. - Four of the selector/storage register inputs are applied to PINS 6, 8, 10, & 16 and the other four are received internally from a decade counter; the outputs are yielded on PINS 18, 14, 12, & 11.

3.2.1.8 Operational amplifier network #1. - The input is applied to PIN 3, and the output is yielded on PIN 5.

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3.2.1.9 Operational amplifier network #2. - The input is applied to PIN 7 and the output is yielded on PIN 9.

3.2.2 Environmental. - The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics. - The physical characteristics shall be as specified on drawing 11745212 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings. - This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11745212 and associated lists.

3.3.2 Standard of Manufacture. - The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship. - The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample. - Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection. - The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination. - The assembly shall undergo first article and counter testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection. - The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment. (see Figure 1)

4.2.1.1 Power Supply. - The power supplies shall be as follows:

a. +10.000 VDC ±1 mVDC, 100 mA or greater and a ripple of less than 10 mVRMS.

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b. +20.0 VDC ±0.1 VDC, 100 mA or greater and a ripple of less than 10 mVRMS.

c. +40.000 VDC ± 1 mVDC, 100 mA or greater and a ripple of less than 10 mVRMS.

d. -10.000 VDC ±1 mVDC, 100 mA or greater and a ripple of less than 10 mVRMS.

e. -20.0 VDC ±0.1 VDC, 100 mA or greater and a ripple of less than 10 mVRMS.

4.2.1.2 Pulse generator. - The generator shall be capable of producing a single 10 nanosecond pulse and a single 10 nanosecond inverse pulse, each shall have an amplitude of 0.0 VDC to 5.0 VDC ±0.1 VDC.

4.2.1.3 Dummy loads. - The dummy loads shall consist of eleven triple (3) - input nand gates, type SN54L10 or their equivalents; and five 10K ohm, ¼ watt resistors. Two of the resistors shall have tolerances of ±5%, and three tolerances of ±.1%.

4.2.1.4 Digital Voltmeter. - The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input impedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance. - Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power Supply. - All requirements shall be met with the following power supplies:

a. +5.000 VDC ± 1 mVDC
b. +15.0 VDC ± 1mVDC
c. +28.000 VDC ±1 mVDC
d. -5.000 VDC ±1 mVDC
e. -15.0 VDC ± 1 mVDC

4.2.2.2 Digital Output. - The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

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4.2.2.3 Test Set-up. - The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.3.1 Voltage divider/Nand gate network test. - When the stated inputs are applied to PINS 39 & 29, output PINS 36 & 37 shall function as shown in Table 1.

TABLE 1

INPUTS		OUTPUTS	
PIN 39 (VDC)	PIN 29	PIN 36	PIN 37
0V ±1 m V	Logic 0	----------	Logic 1
0V ±1 m V	Logic 1	----------	Logic 1
28V ±1 m V	Logic 0	----------	Logic 0
28V ±1 m V	Logic 1	4.23 VDC ±.05 VDC	Logic 1

4.2.2.3.2 Flip-flop/Nand gate network test. - When the stated inputs are applied to input PINS 39, 29, & 32 and the instructions are followed concerning Reset 0 on PIN 23, output PIN 41 shall function as shown in Table 2.

TABLE 2

INPUTS			OUTPUTS	
PIN 39 (VDC)	PIN 29	INSTRUCTIONS	PIN 32	PIN 41
0V ±1 m V	Logic 0	Reset 0 (PIN 23)	Logic 0	Logic 0
28V ±1 mV	Logic 1	Reset 0 (PIN 23)	Logic (1) 0 Logic 1	Logic 1
----------	-------	Reset 0 (PIN 23)	----------	-------
28V ±1 mV	Logic 0	----------	Logic 0	Logic 0
0V ±1 mV	Logic 1	----------	Logic 0	Logic 0

NOTE: (11 Switch from Logic 0 to Logic 1 after Reset 0 (PIN 23) and inputs to PINS 39 & 29 have been applied.

4.2.2.3.3 Nand gate test. - The input to PIN 25 and the output on PIN 13 shall be as shown in Table 3.

TABLE 3

INPUT	OUTPUT
PIN 25	PIN 13
Logic 0	Logic 1
Logic 1	Logic 0

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4.2.2.3.4 Flip flop/Nand gate/Decade counter network test procedure. - Connect output PIN 33 to input PIN 32.

Reset 0 on PIN 23 and Reset 1 on PIN 31, output PIN 33 shall be Logic 1.

Apply a 10 nanosecond Logic 1 pulse to PIN 17, output PIN 33 shall be Logic 1. Again apply a 10microsecond. Logic 1 pulse to PIN 17, output PIN 33 shall be Logic 0. Again apply a 10microsecond. Logic 1 pulse to PIN 17, output PIN 33 shall be Logic 0. Again apply a 10microsecond Logic 1 pulse to PIN 17, output PIN 33 shall be Logic 1.

Output PIN 33 shall remain at Logic 1 regardless of how many Logic 1 pulses are applied to PIN 17.

4.2.2.3.5 4-Bit data selector/Storage register #1 test. - When the inputs are applied as stated, output PINS 22, 24, 26, & 28 shall function as described in the below test procedure.

4-Bit data selector/Storage register #1 test procedure. - Apply Logic 0 to PINS 32, 39, 21, & 19 and Logic 1 to PINS 20 & 27. Output PINS 22 & 26 shall be Logic 0, and PINS 24 & 28 Logic 1.

Apply Logic 0 to PINS 20 & 27, and Logic 1 to PINS 21 & 19. Output PINS 24 & 28 shall be Logic 0 and PINS 22 & 26 shall be Logic 1.

Apply Logic 0 to PINS 21, 20, 19 & 27, and +28.000 VDC ±1m VDC to PIN 39, and Reset 1 on PIN 31. Apply 5 nanoseconds. Logic 1 pulses to PIN 17. Output PINS 24 & 28 shall be Logic 0 and output PINS 22 & 26 shall be Logic 1.

Reset 1 on PIN 31. Output PINS 22 & 26 shall be Logic 0.

Apply 2-10microseconds Logic 1 pulses to PIN 17, output PIN 24 shall be Logic 1.

Reset 1 on PIN 31. Apply 8-10microseconds Logic 1 pulses to PIN 17, output PIN 28 shall be Logic 1.

4.2.2.3.6 4-Bit data selector/Storage register #2 test. - When the inputs are applied as stated, output PINS 18, 14, 12, & 11 shall function as described in the below test procedure.

4-Bit data selector/Storage register #2 test procedure. - Apply Logic 0 to PINS 32, 39, 6, & 10 and Logic 1 to PINS 8 & 16. Output PINS 14 & 11 shall be Logic 0 and PINS 18 & 12 shall be Logic 1.

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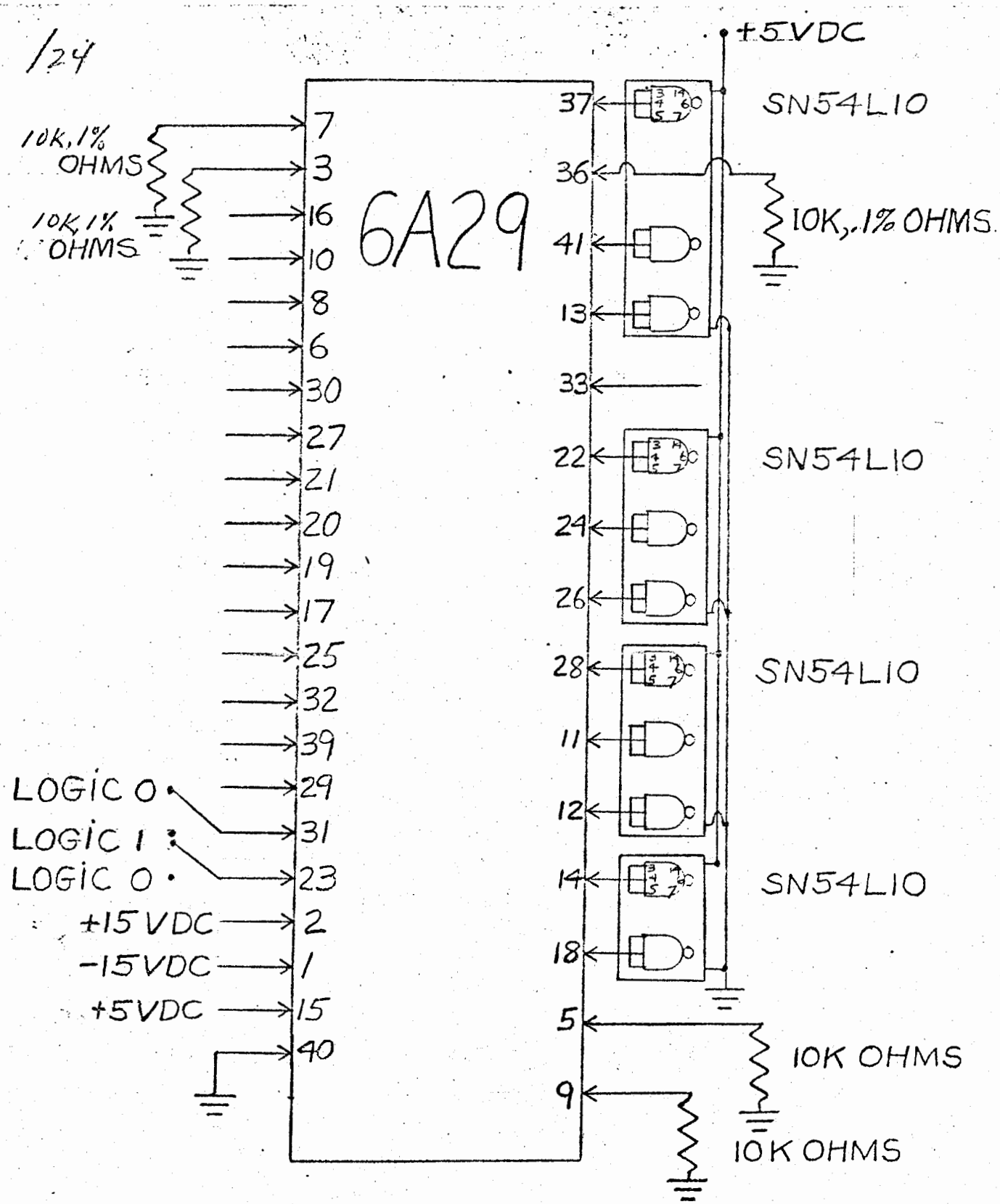


FIGURE 1

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Apply Logic 0 to PINS 8 & 16 and Logic 1 to PINS 6 & 10. Output PINS 18 & 12 shall be Logic 0 and PINS 14 & 11 shall be Logic 1.

Apply Logic 0 to PINS 6, 8, 10, & 16 and 28.000 VDC ±1 VDC to PIN 39, and Reset 1 to PIN 31. Apply 50 10microseconds Logic 1 pulses to PIN 17. Output PINS 18 & 12 shall be Logic 0 and PINS 14 & 11 shall be Logic 1.

Reset 1 at PIN 31. Output PINS 14 & 11 shall be Logic 0.

Apply 20 10microseconds Logic 1 pulses to PIN 17, output PIN 12 shall be Logic 1.

Reset 1 at PIN 31. Apply 80 10microseconds Logic 1 pulses to PIN 17, output PIN 18 shall be Logic 1.

4.2.2.3.7 Operational amplifier network #1 test. - The input to PIN 3, and the output on PIN 5 shall be as shown in Table 4.

TABLE 4

INPUT (VDC) PIN 3	OUTPUT (VDC) PIN 5
0V ±1 m V	0V ±.2V
5V ±1 m V	5V ±.2V
-5V ±1 mV	5V ±.2V

4.2.2.3.8 Operational amplifier network #2 test. - The input to PIN 7 and the output at PIN 9 shall be as shown in Table 5.

TABLE 5

INPUT (VDC) PIN 7	OUTPUT (VDC) PIN 9
0V ±1 m V	0V ±.1V
5V ±1 m V	-5.65V ±.1V
-5V ±1 mV	+5.65V ±.1V

5.1 Packaging. - Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11745212.

6. NOTES

6.1 The notes specified in the general specification MIL-G-48411 are applicable to this specification.

Custodian:

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Preparing Activity:

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Project No. 1240-A619

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